

Redefining 'self': the role of microflora (commensals) mismatch in the development of GvHD after allogeneic stem cell transplantation and some possible remedies

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Summary The discovery of human leukocyte antigen (HLA) molecules and their role in allorecognition has facilitated the initiation of allogeneic stem cell transplantation in human beings. HLA mismatch to a large extent explains the phenomenon of graft rejection and graft versus host disease (GvHD). Incidence of GvHD even in syngeneic transplants suggests a role for extra genetic factors in the causation of GvHD. We hereby propose a hypothesis that the definition of 'self' (in the immunological sense) should be broadened to include both genetically determined molecules (e.g. HLA) and the microbial flora that colonize an individual. This hypothesis explains several observations about GvHD which can not fully be accounted for by the HLA mismatch theory and gives some clues towards circumventing GvHD. © 2001 Harcourt Publishers Ltd

Allogeneic stem cell transplantation is the treatment of choice for the management of various malignant and non-malignant disorders. However, graft versus host disease, one of the major complications of this treatment continues to cause significant morbidity and mortality (1).

During allogeneic transplantation the transfer of immunologically competent cells of the normal donor to the immunologically compromised and genetically different recipient is stated to cause the GvHD. Cytotoxic T lymphocytes have been generally thought to play an important effector role. Immune cells of the donor origin recognize recipient tissues as foreign, and attack and destroy them (2).

Current opinion about the pathogenesis of GvHD involves a mismatch of the major histocompatibility antigens, comprising mainly the human leukocyte antigenic (HLA) system, between the donor and the recipient. However, with a variable incidence ranging from 25–75% in different centers, GvHD has also been observed in recipients of HLA identical grafts (2). This has been attributed to the mismatch of as yet inadequately characterized minor HLA antigens (3). GvHD occurs even in syngeneic transplants, where minor HLA antigens are also expected to be matched between identical twins. These observations argue for a pathogenetic role of non-genetic factors in the development of GvHD.

ROLE OF MICROBIAL FLORA IN THE INDUCTION OF GVHD

Absence of GvHD during studies conducted in germ-free or completely decontaminated rodents demonstrated that the resident intestinal bacterial microflora might contribute towards the pathogenesis of experimental chronic GvHD (4). The absence or complete suppression

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of intestinal bacteria prevented the development of GvHD even in recipient animals of MHC-mismatched transplants (4). Studies in humans have similarly shown that gut decontamination decreases the incidence of GvHD (5).

REDEFINING SELF

'Self' in immunological terms is defined by gene derived molecules comprising primarily the major and minor HLA antigens of an individual. However, we propose that 'self' should actually be defined as the combination of the genetically derived molecules (HLA) and the unique commensal microorganisms that colonize each individual.

We examine some known facts about GvHD which can not be adequately explained by the HLA mismatch theory and show that this can be done using the above hypothesis. In the light of this understanding we propose some measures that may prevent or mitigate GvHD.

Organ predilection for the incidence of GvHD

One of the remarkable features of GvHD is its striking predilection for only some organ systems, namely skin, gut, liver and mucous membranes. This is despite the fact that all tissues of the body have a common genetically derived antigenic structure (HLA molecules) and therefore should be at equal risk of damage from GvHD.

One feature common to all these organ systems except liver is that they are colonized by microbial flora on their surfaces. In fact, skin and gastrointestinal tract are the sites of maximum colonization in the body.

Even within these sites, damage tends to occur at the actual sites of maximum microbial colonization. For example, in the gut, acute GvHD tends to occur frequently and in greater severity in ileum and colon compared to stomach (which is least colonized because of its acidic milieu) (6,7). Similarly, ocular involvement in GvHD is in the form of conjunctivitis rather than uveitis, in contrast to other autoimmune disorders where uveitis is quite prominent (6). Again conjunctiva is colonized by microbes unlike uvea.

Liver is the only organ which is not colonized by microflora and is prominently involved in GvHD. However, liver is the organ which first filters the portal venous blood coming from the heavily colonized GIT. It is known that microbes or their antigenic components reach the liver in portal blood through the phenomenon of 'microbial translocation' which are trapped by kupffer cells and thereafter remain in the liver (8). This phenomenon is especially known to be increased with malignancy (9) and with the use of chemotherapeutic drugs (10). Thus, even though liver is not colonized in the classical sense, it does contain significant amounts of microbial

antigens. In this context, it is interesting to note that biliary cirrhosis for which a microbial etiology has been postulated (11,12) closely resembles liver involvement of chronic GvHD (13), clinically and morphologically.

The arguments advanced thus far strongly favour microbial involvement in the pathogenesis of GvHD. In the next sections, we argue that it is not merely the microbial colonization of organ systems, but the mismatch of colonizing organisms between the host and the recipient which may be important in the pathogenesis of GvHD.

Influence of age on the development of GvHD

Increasing age has been known to significantly increase the incidence of GvHD in HLA-matched sibling transplants. Therefore, Allogeneic Stem Cell Transplantation (ASCT) is avoided by many centers, for patients older than 50 years.

The role of age in the incidence of GvHD cannot be explained by mismatch of major or minor HLA molecules which are age independent. Even microbial colonization by itself can not explain the age dependence of GvHD, since individuals of all ages continue to be colonized. The alternative mechanism could point towards food and environment influencing the microbial colonization pattern in individuals with advancing age (14,15). Younger donor-recipient pairs are likely to share the same food habits and environments compared to older pairs. The latter tend to live separately and acquire new food habits due to various social, occupational and matrimonial reasons. This makes it likely that with increasing age sibling pairs harbour different sets of commensal flora, thereby increasing the incidence of GvHD (due to altered 'self' according to our hypothesis).

Moreover, unrelated but HLA matched donor-recipient pairs are unlikely to have the same commensal pattern, irrespective of age. It might be predicted that in matched unrelated ASCT, the incidence of GvHD should not correlate with age. This has indeed been confirmed in a recent study reported from a prestigious center (16). The incidence of GvHD in this situation is high even in younger patients possibly because of mismatched flora, due to above reasons.

Influence of T cell subtypes on the incidence of GvHD

Helper T cell (CD 4) frequencies directly correlate with the occurrence of GvHD (17). Helper T cells respond to exogenous antigens in the context of Class II HLA molecules. This is in contrast to suppressor/cytotoxic T (CD 8) cells which respond to endogenous antigens presented with class I HLA molecules. The most obvious source of exogenous antigens is the microbial flora at the site of

development of GvHD, thereby explaining the correlation between incidence of GvHD and helper T cell frequencies.

Role of viruses on the incidence of GvHD

The incidence of GvHD increases with increasing infection rates by herpes viruses (e.g. cytomegalo virus) in both donors and recipients (18,19). This phenomenon can be explained by 'self' being defined by both HLA molecules and the viral antigens in these individuals leading to potential mismatches.

SOME PRACTICAL MEASURES WORTH EXPLORING TO MITIGATE GVHD

We propose that efforts to make the microbial flora as similar as possible between the donor-recipient pairs might decrease both the incidence and severity of GvHD. This could be achieved by enforcing similarity of diet and environment between the selected pairs for a certain period prior to transplantation. Alternatively, more direct methods like bathing in shared water, wearing shared clothes, etc. could be adopted.

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